

Recommendation and Large Models in Research and Industry: A Comprehensive Survey from Neural Ranking to LLM-Enhanced, Generative, and Agentic Recommenders

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Abstract

Large language models, multimodal foundation models, large sequential transducers, and generative retrieval models are reshaping recommender systems. Early work asked whether language models could act as zero-shot recommenders or explanation generators; recent work asks how to build foundation-scale retrieval, ranking, re-ranking, intent reasoning, conversational recommendation, and user simulation systems that survive production constraints. This survey provides a broad view of the field from approximately 2016 to July 2026. It covers classical deep recommenders, transformer sequential recommenders, pre-trained language models for recommendation, LLM-as-feature and LLM-as-ranker methods, semantic-ID and generative retrieval, multimodal and knowledge-enhanced systems, agentic recommenders, LLM-based evaluation, fairness and safety, and industrial deployments. We distinguish three evidence tiers: (1) public industrial systems with online A/B results, (2) industrial or production-oriented systems with deployment claims or offline/internal evaluation but limited public A/B disclosure, and (3) academic systems that define the technical frontier but are not yet proven in large-scale online traffic. Across the literature, the central industrial lesson is that large models rarely replace the whole recommender stack. The most successful systems transform large-model capability into recommender-native artifacts: semantic IDs, compact user-intent memories, generated candidates, knowledge graph edges, multimodal tags, reward features, or calibrated ranking signals. We conclude with a deployment playbook, a risk analysis, and a paper map covering more than two hundred related works and systems.

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1 Scope, Evidence Tiers, and Survey Method

The phrase “recommendation plus large models” now covers several different technologies. It may refer to a BERT-style encoder used to represent news articles, a T5-style text-to-text model such as P5, a LLaMA-style model tuned for binary preference prediction, a multimodal LLM that extracts product attributes, a semantic-ID generator for retrieval, a trillion-parameter sequential transducer for ranking, or an agent that plans multi-step interaction with a user. These systems have different failure modes and very different deployment costs. Treating them as one category hides the engineering decisions that matter most.

This survey uses three evidence tiers.

1. **Tier 1: public online evidence.** The source reports production A/B or live experiment results. These cases are the strongest evidence for industrial value, although the metrics and baselines are still not directly comparable.
2. **Tier 2: production-oriented but incomplete online disclosure.** The source is industrial, uses real logs, or describes deployment architecture, but does not disclose full online A/B results. These works are valuable for architecture and systems lessons but should not be counted as proven online wins.
3. **Tier 3: academic and pre-production research.** The work advances modeling, alignment, evaluation, robustness, fairness, or data methods, usually on public datasets. These papers define the technical frontier and frequently precede later deployment, but their offline gains should not be confused with production impact.

The time span is broad. We start from neural collaborative filtering, Wide&Deep, YouTube DNN, DIN, DIEN, SASRec, BERT4Rec, and DLRM, because these systems explain why the industrial recommender stack looks the way it does. We then follow the shift to pre-trained language models and LLMs: P5, M6-Rec, TALLRec, Chat-REC, InstructRec, ReLLa, CoLLM, LLaRA, A-LLMRec, TIGER, HSTU, PLUM, SIGMA, RecGPT, and many others. The latest papers included in the paper map are from July 2026, including LBR for length bias, multimodal memory-enhanced agent collaboration, LLM-based offline evaluation, connected-TV agentic recommendation, and controllability-centric evaluation.

2 Historical Background: Why Recommendation Did Not Simply Become NLP

2.1 The industrial recommendation stack

Industrial recommenders are multi-stage systems. Retrieval or candidate generation narrows a corpus of millions or billions of items. Pre-ranking and ranking score candidates under tight latency

budgets. Re-ranking or slate optimization balances diversity, novelty, fairness, revenue, inventory, creator or merchant health, and policy constraints. Feature stores, streaming logs, online learning, ANN indexes, experimentation platforms, and guardrails are as important as the neural model itself.

This stack grew around sparse IDs, dense features, and behavior logs. Wide&Deep and DeepFM made cross features and dense embeddings practical for CTR prediction. YouTube DNN candidate generation showed how large-scale retrieval could be framed as multiclass classification over a huge corpus. DIN and DIEN emphasized target attention and evolving user interest. SASRec and BERT4Rec brought transformer sequence modeling to sequential recommendation. DLRM and its industrial descendants scaled sparse embedding tables and feature interaction networks. These models are not obsolete. They remain the serving substrate into which many LLM-based methods are inserted.

2.2 What language models add

Language models add four capabilities that classical ID recommenders do not naturally have.

- **Semantic abstraction.** A product title, review, query, image caption, video transcript, or user self-description can be mapped into a space where cold-start and cross-domain transfer become easier.
- **Instruction following.** Recommendation tasks can be expressed as prompts: predict whether a user will like an item, rank a candidate list, explain a recommendation, ask a clarifying question, or generate a search query.
- **World knowledge and reasoning.** LLMs can infer seasonal intent, substitute/complement relations, event-driven preferences, and high-level attributes not explicitly present in interaction logs.
- **Text generation and interaction.** LLMs can produce explanations, summaries, conversational turns, synthetic user profiles, and natural-language critiques of recommendation results.

At the same time, recommenders impose requirements that general LLMs do not satisfy by default: calibrated scores, low latency, huge catalogs, frequent item churn, counterfactual evaluation, business constraints, and stable online experimentation. Much of the literature can be read as attempts to reconcile these two worlds.

3 A Taxonomy of Large-Model Recommendation

3.1 LLM as feature engineer

The simplest and most deployable role is feature engineering. A frozen or lightly tuned LLM extracts item tags, user profiles, intent summaries, review aspects, query rewrites, graph edges, or synthetic samples. Examples include TagGPT, LLM4KGC, KAR, HKFR, RLMRec, LLMRec with graph augmentation, CUP, SINGLE, LLMHG, Llama4Rec, ONCE, MINT, RecPrompt, BEQUE, and Agent4Ranking. The model is not asked to score every candidate online; it generates artifacts that can be cached and consumed by conventional retrieval or ranking systems.

This is why LLM-as-enhancer is often industrially attractive. It localizes risk. If the generated attribute is noisy, the downstream ranker can learn to ignore it; if the LLM is expensive, generation can run offline; if a policy rule changes, the feature pipeline can be audited separately.

3.2 LLM as encoder

Pre-trained language models can encode text-rich items and user histories. This line starts before the current LLM wave: U-BERT, UNBERT, PLM-NR, Pyramid-ERNIE, ERNIE-RS, CTR-BERT,

Tiny-NewsRec, RecFormer, UniSRec, VQ-Rec, MISSRec, PMMRec, and many news, job, software, and tourism recommenders. Later work adds larger LLM embeddings, instruction-tuned encoders, or adapters.

Encoder approaches fit industrial stacks because embeddings can be precomputed, indexed, distilled, and combined with ID embeddings. Their limitation is that they often capture content similarity better than collaborative preference. This is the semantic-collaborative gap that motivates CoLLM, LLaRA, A-LLMRec, LLM2BERT4Rec, LC-Rec, ControlRec, and related alignment methods.

3.3 LLM as scorer or ranker

Another line turns recommendation into language classification or ranking. TALLRec expresses user history and a candidate item as an instruction and asks the LLM to answer yes/no. InstructRec and RecRanker use instruction tuning. SetwiseRank and listwise LLM rankers use comparative prompting to rank candidate sets. ReLLa retrieves relevant behavior snippets before prompting an LLM, while LlamaRec uses a two-stage retrieval-then-rank design. These systems are attractive for controllability and cold-start reasoning, but raw LLM scoring is usually too slow and poorly calibrated for large industrial ranking.

3.4 LLM as generator

Generative recommendation treats the output item, item ID, semantic ID, query, explanation, or slate as a generated sequence. P5 and M6-Rec framed multiple recommendation tasks as language processing. GPT4Rec, GenRec, BIGRec, PALR, LightLM, LLaRA, LC-Rec, IDGenRec, TIGER, and many semantic-ID systems extend this idea. The key design question is the output vocabulary: textual item titles are human-readable but ambiguous and hard to ground; numeric IDs are precise but semantically empty; semantic IDs offer a compact compromise.

3.5 Large recommendation models and sequential transducers

Not all large recommendation models are LLMs. Meta’s HSTU-based Generative Recommenders reformulate industrial recommendation as sequential transduction over user actions and scale to 1.5 trillion parameters, reporting a 12.4% online A/B improvement and power-law scaling with compute [18]. Meituan’s MTFM builds a multi-scenario recommendation foundation model with heterogeneous tokens [21]. These models borrow from the language-model scaling playbook but are trained on action streams, sparse IDs, and recommender objectives.

3.6 Agentic and conversational recommenders

Agentic recommenders use LLMs for planning, tool use, memory, self-reflection, or multi-agent collaboration. Chat-REC and conversational systems explore interactive recommendation. MACRec, AgentCF, iAgent, RecMind, RecGPT-V2, MMEACR, LLM-as-a-judge slate recommenders, and connected-TV agentic systems extend this direction. The promise is richer preference elicitation and better explanations. The industrial challenge is predictable behavior, latency, user safety, and evaluation.

4 Core Academic Milestones

4.1 P5 and the text-to-text unification idea

P5, or Pretrain, Personalized Prompt, and Predict, was a milestone because it made the argument that recommendation tasks can be unified as text-to-text learning [7]. Rating prediction, sequential recommendation, explanation generation, review summarization, and direct recommendation can

share the same language-model objective. P5 did not solve industrial serving, but it changed how researchers thought about task unification.

4.2 M6-Rec and open-ended industrial pretraining

M6-Rec extended a large generative pre-trained language model to recommendation and argued that industrial recommenders need open-ended support for multiple domains, tasks, and modalities [8]. It converted recommendation tasks into language understanding or generation tasks and emphasized sample-efficient downstream adaptation. It is an important bridge between academic LLM4Rec and industrial foundation models.

4.3 TALLRec and instruction tuning for preference prediction

TALLRec demonstrated that LLaMA-style instruction tuning with LoRA can align an LLM to recommendation data [9]. Its yes/no formulation is simple but influential. It also exposed a recurring limitation: language-model semantics and collaborative preference are not the same thing. Later methods improve this with collaborative embeddings, retrieval augmentation, rationale distillation, or semantic IDs.

4.4 TIGER and semantic-ID generative retrieval

TIGER, “Recommender Systems with Generative Retrieval,” reframed retrieval as generating item identifiers [14]. Instead of nearest-neighbor search over embeddings, the model decodes a semantic ID learned from item content. This idea inspired a large family: P5-ID, LC-Rec, IDGenRec, LETTER, RPG, differentiable semantic IDs, variable-length semantic IDs, SIDInspector, PLUM, SIGMA, AKT-Rec, and Gryphon. The central questions are collision, validity, calibration, cold-start generalization, and efficient constrained decoding.

4.5 Collaborative alignment methods

CoLLM, LLaRA, A-LLMRec, LC-Rec, ControlRec, CLLM4Rec, and related methods try to inject collaborative signals into language models. They differ in mechanism: projection of collaborative embeddings into token space, adapters, item token learning, contrastive alignment, or two-stage grounding. This family exists because pure text semantics often fails on behavior-driven preference.

4.6 RAG and long-history understanding

ReLLa and RALLRec show that LLMs need retrieval to handle lifelong behavior. Long user histories contain both signal and noise. A retrieval step selects relevant behaviors before prompting or encoding. AIR, RecGPT-V2, and RecGPT-Mobile industrialize a similar insight: compress the user’s history into task-relevant intent representations before ranking.

4.7 Evaluation, fairness, and robustness

Recent work studies whether LLM recommenders are stable, fair, calibrated, and controllable. STELLA argues that LLM recommenders can be unstable. FaiRLLM, RECLLM, fairness surveys, provider fairness studies, job recommendation bias analyses, and demographic bias studies show that LLMs inherit and sometimes amplify social and popularity biases. LBR, submitted in July 2026, targets length bias in LLM-based recommendation. LLM-based offline evaluation and controllability-centric agent evaluation show that evaluation itself is changing.

5 Industrial Systems with Public Online Evidence

Table 1 summarizes industrial cases where public sources disclose online A/B or live-experiment metrics. These results should not be compared as if they were from one benchmark. They differ in traffic, baseline maturity, product surface, metric definitions, and time horizon. Still, they show which patterns have crossed the production barrier.

Table 1: Public industrial online evidence for large-model recommendation.

System	Platform	Large-model role	Public online result	Source
HSTU Generative Recommenders	/ Meta large internet platform	Sequential transduction architecture for retrieval/ranking over action streams.	1.5T-parameter GRs report +12.4% online A/B topline improvement; HSTU is 5.3x–15.2x faster than FlashAttention2 Transformers on 8192-length sequences.	[18]
ReLand	Alipay	LLM insights injected through a controllable reasoning pool.	Deployed since Jan. 2024; +3.19% CTR and +1.08% CVR .	[19]
LLM-KERec	Alipay	Inferential knowledge graph built with LLM assistance.	One-month A/B with 10% traffic: +6.24% conversion, +6.45% GMV , +10.07% conversion across three scenarios.	[20]
MTFM	Meituan	Multi-scenario recommendation foundation model.	SQS: +1.89% CTR , +2.46% UV_CTCVR , +2.98% orders , -5 ms . PHF: +1.53% CTR , +1.03% UV_CTCVR , +1.45% orders , -6 ms .	[21]
SIGMA	AliExpress	Semantic-grounded instruction-driven generative multi-task recommender.	Two-week A/B with 5% traffic: +2.80% order volume , +3.84% conversion , +7.84% GMV , +2.47% purchased category breadth .	[22]
SORT-Gen	Taobao	Generative list-level multi-objective re-ranking.	Two-week A/B on Baiyibutie: +4.13% CLICK and +8.10% GMV .	[23]
AIR	Kuaishou	Offline LLM atomic intents with online target-aware retrieval.	5.08% traffic: +1.043% paid orders , +3.446% GMV , +3.662% GPM , +1.254% OPM ; about 400x throughput gain over direct LLM invocation.	[24]
Taiji	Kuaishou Ads	LLM-as-enhancer with preference reasoning and Pareto reward alignment.	One-week A/B, 10%/10% split: +2.83% ADVV , +3.30% revenue ; long-tail users +5.26% ADVV , +5.32% revenue .	[25]
PLUM	YouTube	Gemini-family pre-trained LM adapted to semantic-ID generative retrieval.	Live experiments adding PLUM over LEM+: LFV +0.76% panel CTR , +0.80% views ; Shorts +4.96% panel CTR , +0.39% views ; satisfaction positive.	[26]
AKT-Rec	Tmall	MLLM/LLM semantic IDs with asymmetric head-to-tail transfer.	Two-week A/B, 10%/10% split: +2.73% clicks , +2.76% CTR , +1.7% CTCVR , +3.47% GMV .	[27]

System	Platform	Large-model role	Public online result	Source
RecGPT	Taobao	Hundred-billion-scale intent-centric reasoning model.	Public summaries report online +9.47% IPV and +6.33% CTR , with diversity and merchant-side benefits.	[28]
RecGPT-V2	Taobao	Hierarchical multi-agent intent reasoning and agentic judge/reward.	Two-week A/B against RecGPT-V1. Item scenario reports +3.64% IPV , +3.01% CTR , +3.39% GMV , +3.47% ATC , +11.46% NER .	[29]

6 Industrial and Production-Oriented Systems without Full Public A/B

Many important industrial systems do not disclose public A/B metrics. They still shape the field because they describe architecture, training pipelines, system bottlenecks, and deployment assumptions.

- **M6-Rec and M6-like e-commerce foundation models** argue for unified generative modeling across product, content, and user-generated domains.
- **BEQUE and Taobao query rewriting** show how LLMs can generate long-tail query rewrites for search/recommendation, often as offline or nearline expansions.
- **RecGPT-Mobile** moves a lightweight intent agent onto the user device, preserving privacy and immediacy while leaving large retrieval/ranking in the cloud.
- **Large Language Model as Universal Retriever** explores whether one LLM-based retriever can serve multiple industrial retrieval objectives.
- **Gryphon** adds item-level scoring to semantic-ID generation in an industrial music service, addressing sequence-score calibration and SID collisions.
- **RecBase, OneMall, GPR, RankGR, and related large recommender models** aim to pretrain or align one model across multiple recommendation surfaces, though public online evidence varies.
- **Meta GEM and ads foundation-model discussions** suggest a move toward shared ads recommendation backbones and accelerated experimentation, but public third-party blog claims should be treated as contextual unless paired with primary technical reports.
- **Google/YouTube product and research announcements** around Gemini and creator/advertiser matching show how multimodal foundation models are entering adjacent recommendation-like matching tasks; PLUM remains the clearest public technical evidence for YouTube recommendation A/B impact.

7 Technical Themes

7.1 Semantic IDs and the discrete bottleneck

Semantic IDs solve one problem while creating several new ones. They compress rich item semantics into a short discrete code that can be generated by an autoregressive model. This makes retrieval look like language decoding and enables zero-shot or cold-start generalization when the ID is tied to content. But codes can collide, become unbalanced, overfit to modality artifacts, or produce invalid

outputs. Recent work responds with residual quantization, learnable item tokenization, differentiable semantic IDs, variable-length IDs, unordered long IDs, parallel decoding, qualification-aware collision handling, adaptive collision handling, and diagnostic resources such as SIDInspector.

7.2 Collaborative signal alignment

LLMs know language; recommenders know behavior. Alignment methods try to prevent one from washing out the other. Some methods project collaborative embeddings into language-model token space. Others train adapters or LoRA modules. Some use contrastive objectives to align item texts with co-clicks. Semantic-ID methods often build the ID from both content and collaborative signals. Industrial systems such as PLUM and AKT-Rec show that this alignment is not optional: content-only semantics is rarely enough for high-stakes recommendation.

7.3 Long histories and memory

Large context windows do not automatically solve lifelong behavior modeling. User histories include stale, contradictory, accidental, and context-dependent interactions. ReLLa, RALLRec, AIR, RecGPT-V2, RecGPT-Mobile, HLLM-style hierarchical methods, and MMEACR-style multimodal memory agents all reflect the same insight: retrieve and compress the relevant parts of history before the large model or ranker sees them.

7.4 Reranking, slate optimization, and multi-objective learning

Large models are useful beyond top-1 relevance. SORT-Gen models list-level multi-objective reranking. LLM-based diversity re-ranking and explanation-aware re-ranking optimize properties of the whole slate. RecGPT-V2 includes novelty exposure rate. SIGMA includes purchased category breadth. Taiji balances semantic and collaborative reward. The research frontier is moving from item scoring toward controllable slate generation under business constraints.

7.5 Multimodal recommendation

Modern items are multimodal: product images, titles, reviews, videos, audio, transcripts, comments, creator metadata, and live-stream context. VIP5, MM-Rec, MISSRec, PMMRec, LLMs in multimodal recommender surveys, AKT-Rec, PLUM, and MMEACR all show different ways to combine modalities. The hard part is not simply using a vision-language model; it is deciding which modality should dominate under which user intent and how to keep multimodal features fresh and policy-safe.

7.6 Agents, interaction, and user simulators

Agentic recommenders can ask questions, revise assumptions, call tools, simulate users, critique slates, and maintain memory. Agent4Rec, RecMind, MACRec, AgentCF, iAgent, MMEACR, LLM-based group recommenders, connected-TV agents, and LLM-as-a-judge evaluation systems show this shift. The main risk is that agents are harder to evaluate than rankers. A ranker returns a score; an agent may change the user’s preference statement, ask a question, or hallucinate a constraint.

8 Problem-Oriented Reading Guide

The literature is easier to navigate when organized by the problem a practitioner or researcher is trying to solve. The same model family can be appropriate in one setting and wasteful in another.

8.1 Cold start and long tail

Cold-start and long-tail settings are the most natural entry point for LLMs. Traditional ID embeddings are weak when an item has few exposures or a user has little history. Content-rich LLM representations, semantic IDs, and multimodal encoders can transfer information from item titles, product images, descriptions, reviews, transcripts, and co-occurring entities. In this setting, the most relevant works include P5, M6-Rec, TIGER, UniSRec, VQ-Rec, MISSRec, PMMRec, AKT-Rec, PLUM, GenCDR, and semantic-ID diagnostics. The industrial lesson is to avoid content-only features as a final answer: cold-start content semantics should be fused with whatever collaborative evidence is available, even if it is weak. AKT-Rec is important because it explicitly avoids harming head IDs while helping tail IDs; this is a common production concern because improving the tail at the expense of head traffic can reduce total business value.

8.2 Cross-domain transfer

Cross-domain recommendation benefits from LLMs when domains have semantic overlap but sparse direct interactions. Examples include content-to-commerce, search-to-recommendation, short-video-to-product, and query-to-item transfer. AIR, SIGMA, MTFM, S&R Foundation, Uni-CTR, PCDR, and GenCDR illustrate different solutions. A practical design choice is whether to align domains into a shared embedding space, a shared semantic-ID vocabulary, a shared user-intent memory, or a shared multi-scenario transformer. Shared representations improve transfer, but can also create negative transfer when domains optimize different objectives. Production systems therefore often use mixture, gating, adapters, or scenario-specific subgraphs rather than a single unconditioned representation.

8.3 Text-rich and multimodal items

News, e-commerce, local services, jobs, short videos, music, books, and creator/advertiser matching are text-rich or multimodal. PLM-NR, MM-Rec, VIP5, MISSRec, PMMRec, PLUM, AKT-Rec, and MMEACR are representative. For these scenarios, LLMs and MLLMs are valuable not only because they produce better item embeddings, but because they expose interpretable attributes: topic, brand, style, seasonality, audience, intent, safety category, and substitutability. This interpretability matters in industry because recommendation teams need debugging handles. A black-box embedding may improve NDCG; a generated attribute can explain why a model started over-serving a trend or under-serving a merchant class.

8.4 Ads and commercial recommendation

Ads recommendation has stronger calibration, auction, and multi-stakeholder constraints than many organic recommendation tasks. A ranking model must estimate action probabilities and values, not merely choose semantically plausible items. This is why LLMs appear first as feature producers or alignment modules in ads systems. Taiji reports ADVV and revenue gains, while HSTU and GEM-like discussions point toward large shared backbones for ads ranking. In ads, an LLM-generated candidate or explanation is insufficient unless it can be tied to calibrated advertiser value, budget pacing, policy constraints, and user experience guardrails. The right question is not whether an LLM can understand an ad, but whether its semantic signal improves calibrated value prediction without destabilizing the marketplace.

8.5 Short-video and feed recommendation

Short-video and feed systems combine rapid trends, multimodal content, creator ecosystems, long user histories, and high feedback volume. HSTU, PLUM, AIR, RecGPT, RecGPT-Mobile, and related systems are important here. These systems show two opposite but compatible directions: very large action-sequence models that learn directly from behavior, and semantic-intent systems

that compress content and context. The practical architecture may use both: a high-throughput action model for the main ranking signal and LLM-derived intent or content features for cold-start, diversity, explanation, or cross-domain transfer.

8.6 Conversational and agentic recommendation

Conversational recommendation is not only “chat plus ranking.” A conversational system must decide when to ask, when to recommend, how to remember, how to recover from wrong assumptions, and how to ground recommendations in available inventory. Chat-REC, RecMind, MACRec, AgentCF, iAgent, MMEACR, RecGPT-V2, and CTV agentic recommendation illustrate the design space. For production, the hardest issue is evaluation. An agent can produce a fluent dialogue while steering the user toward unavailable or unsafe items. Grounding, tool constraints, and audit logs are therefore as important as model quality.

8.7 Evaluation and user simulation

LLM-as-judge and user-simulation systems are attractive because online A/B tests are expensive. But they should be treated as pre-A/B filters, not replacements for real traffic. A simulator can test whether a system follows constraints, whether an explanation is coherent, whether generated IDs are valid, and whether a slate satisfies a known preference profile. It cannot reliably estimate long-term retention, marketplace dynamics, creator incentives, or fatigue without calibration against real user behavior. The best use of LLM evaluation is layered: offline labels and counterfactual estimators, LLM-based diagnostic judgments, small-scale human review, shadow traffic, then controlled online experiments.

9 Deep Dives on Representative Paradigms

9.1 Instruction-tuned LLM recommenders

Instruction-tuned recommenders such as TALLRec and InstructRec are conceptually simple: convert recommendation data into instruction-response pairs and fine-tune an LLM. Their strength is flexibility. The same model can answer whether a user likes an item, explain the choice, or adapt to a natural-language constraint. Their weakness is that recommendation labels are not naturally language labels. A yes/no answer may be easy to train but hard to calibrate as CTR or CVR. The candidate set may also be artificially small in offline experiments. In production, instruction-tuned LLM rankers are more plausible as rerankers, weak-user specialists, explanation modules, or feature generators than as the primary scorer for millions of candidates.

This paradigm is still important because it established the data interface between recommender logs and LLM training. Once user histories and item metadata are serialized into instructions, one can add rationales, retrieved behaviors, collaborative embeddings, business constraints, or safety rules. Much of the later literature can be understood as improving the naive instruction template: better grounding, better collaborative alignment, better retrieval, better output constraints, or better distillation.

9.2 Semantic-ID generative retrieval

Semantic-ID retrieval is one of the most distinctive LLM-era ideas. It changes the retrieval problem from “score all items or search nearest neighbors” to “generate a discrete code that maps to items.” This creates an elegant connection to language modeling but raises several design choices.

First, how should IDs be constructed? Content-only IDs generalize to new items but may ignore collaborative behavior. Collaborative IDs capture behavior but struggle with cold start. Hybrid IDs try to use both. Second, should IDs be fixed or learned end-to-end? Fixed IDs stabilize the target vocabulary, while differentiable or learnable IDs can adapt to task objectives. Third, how

should collisions be handled? A collision can be useful if it groups substitutes, but harmful if it merges unrelated items. Fourth, how should decoding be constrained? Unconstrained generation can produce invalid IDs; prefix trees, sparse transition matrices, or post-verification are needed at scale. Fifth, how should generated IDs be scored? Accumulated token likelihood may not equal item relevance, motivating systems such as Gryphon that add item-level scoring.

The industrial significance of semantic IDs is that they make language-model-style generation compatible with retrieval infrastructure. PLUM demonstrates this at YouTube scale; SIGMA and AKT-Rec show related ideas in e-commerce. The remaining question is whether semantic-ID generation can consistently beat strong approximate nearest-neighbor retrieval after accounting for freshness, invalid outputs, and serving cost.

9.3 LLM-as-enhancer and semantic feature stores

LLM-as-enhancer systems are likely to remain the most common industrial pattern. The LLM produces a semantic artifact; a conventional recommender consumes it. This pattern includes user profiles, item attributes, query rewrites, graph edges, atomic intents, explanations, reward features, and semantic IDs. It works because it respects the existing production stack. Feature stores, rankers, monitors, and A/B platforms do not need to be redesigned around free-form generation.

However, LLM-generated artifacts require their own lifecycle. They need schema definitions, freshness policies, validation rules, drift monitors, backfills, and rollback mechanisms. If a product taxonomy changes, a generated tag can become stale. If a prompt changes, feature distributions can shift. If the LLM provider changes, downstream model calibration can move. Treating semantic artifacts as ordinary features is not enough; they are generated features with prompt and model provenance.

9.4 Recommendation-native foundation models

HSTU and MTFM show a different route: instead of adapting a general LLM, build a large model around recommendation data. This route is attractive when the platform has massive action logs and enough infrastructure to train and serve very large models. It avoids some semantic-collaborative mismatch because the model is trained directly on behavior. It also preserves the possibility of scaling laws over recommendation compute.

The cost is engineering complexity. Recommendation-native foundation models must handle high-cardinality sparse features, non-stationary item vocabularies, streaming updates, long histories, multi-objective labels, and low-latency serving. HSTU introduces architectural changes and inference amortization; MTFM uses heterogeneous tokens and scenario-specific subgraphs. These models are closer to building a new recommender platform than adding an LLM component.

9.5 Agentic recommenders

Agentic recommenders are appealing because they can represent a user’s task, not only the user’s historical clicks. A shopping user may say, “I need a gift for a colleague under a budget,” and an agent can ask clarifying questions, retrieve candidates, reason about constraints, and explain trade-offs. RecGPT-V2 brings this idea into Taobao-style feed recommendation through hierarchical expert agents. MMEACR and connected-TV agents show how memory and multimodal context can be added.

The main research challenge is that agentic recommendation blurs the boundary between recommendation, search, dialogue, and decision support. The main industrial challenge is safety. An agent that interacts with users can manipulate preferences, over-personalize, hallucinate availability, reveal private inferences, or create inconsistent experiences. Agentic systems need grounded tools, bounded action spaces, conversation policies, and evaluation beyond ranking metrics.

10 How to Read Blogs, Technical Reports, and Vendor Claims

The user specifically asked to include official technology blogs such as Google or Meta where useful. These sources are valuable but must be read differently from peer-reviewed or arXiv papers. This survey uses four evidence grades.

1. **Primary technical paper with online metrics.** Example: HSTU, PLUM, MTFM, SIGMA, AIR, Taiji, AKT-Rec. These can support claims about online gains, while still requiring attention to baseline and metric definitions.
2. **Primary technical report or product paper without complete online disclosure.** Example: some RecGPT, RecGPT-Mobile, large retriever, or foundation-model reports. These can support architecture claims but not necessarily quantified business impact.
3. **Official company blog.** Useful for product context, deployment framing, and strategic direction. Blogs often omit baselines, traffic splits, statistical tests, and guardrails, so they should not be the sole source for precise A/B claims unless they provide enough experimental detail.
4. **Third-party blog or commentary.** Useful for interpretation and discovery, but claims must be traced back to primary sources before being treated as evidence. For example, third-party posts about YouTube semantic IDs or Meta GEM help identify trends, but the survey should ground quantitative claims in PLUM or HSTU-style primary documents.

This evidence discipline matters because recommendation is unusually sensitive to metrics. A 0.5% lift on a huge surface can be valuable; a 5% lift on a weak offline subset may disappear online. Without traffic allocation, duration, baseline, and guardrails, a headline number is not enough.

11 Why Many Academic LLM4Rec Methods Do Not Directly Land Online

Many academic LLM recommender papers are useful but not deployable as written. The gap is not a lack of creativity; it is a mismatch between benchmark assumptions and production constraints.

Candidate-set size. Many LLM ranker experiments rank tens or hundreds of candidates. Industrial retrieval and ranking may consider millions or billions of items across stages. Methods that call an LLM per candidate rarely survive this scale.

Label semantics. Offline datasets often treat observed interactions as positives and sampled items as negatives. In production, unobserved items may be unseen, unavailable, policy-blocked, already consumed, or simply not exposed. LLM evaluation can make this worse if it judges semantic plausibility rather than actual user response.

Temporal leakage. LLMs are pretrained on public corpora and may know item popularity or descriptions from after the training period of a benchmark. Recommendation benchmarks need careful temporal splits and leakage checks.

Catalog grounding. A generated title is not a recommendation unless it maps to an available item. Academic text-generation metrics often ignore inventory, duplicates, seller constraints, price changes, and regional availability.

Latency and cost. A method that improves NDCG with a 7B model may be unusable if it adds tens of milliseconds to a critical ranking path. Industrial papers that report latency or throughput are therefore especially valuable.

Guardrails. CTR or NDCG can improve while diversity, satisfaction, fairness, creator exposure, or policy safety deteriorates. Real deployments need multi-metric scorecards.

Operational drift. Prompts, LLM versions, taxonomy definitions, item catalogs, and user behavior all drift. A generated feature pipeline must be monitored like a model, not like static metadata.

12 Recommended Evaluation Protocol

A robust evaluation protocol for LLM-enhanced recommendation should be staged.

1. **Artifact validation.** Check generated tags, semantic IDs, query rewrites, and intent summaries for validity, coverage, safety, and drift.
2. **Offline ranking evaluation.** Use temporal splits, multiple negative sampling schemes, and segment-level metrics for cold-start, long-tail, new users, and heavy users.
3. **Counterfactual and replay checks.** Use inverse propensity or doubly robust estimators where logging propensities are available; avoid treating missing exposure as negative preference.
4. **LLM diagnostic evaluation.** Use LLM judges only for bounded checks such as explanation coherence, constraint satisfaction, and semantic coverage. Calibrate these judgments against human and behavioral data.
5. **Shadow deployment.** Run the model on live traffic without affecting exposure to measure latency, invalid outputs, feature drift, and candidate overlap.
6. **Online A/B.** Start with small traffic, clear guardrails, and segment analysis. Include business metrics, user metrics, ecosystem metrics, and system metrics.
7. **Long-term holdout.** Measure retention, fatigue, diversity, creator/merchant health, and policy outcomes over longer windows.

The key principle is that LLM-specific evaluation should add gates, not remove the need for online experimentation. A recommender changes user behavior and ecosystem incentives; only live experiments can measure the full feedback loop.

13 Evaluation: Offline, Online, and LLM-as-Judge

Offline recommendation evaluation is already difficult because missing labels are not negative labels. LLMs add new complications: prompt order sensitivity, output-format failures, invalid item generation, position bias, length bias, popularity bias, hallucination, and preference drift. Recent work proposes LLM-based preference induction for offline top-N evaluation, A/B agents, user simulators, controllability-centric evaluation with collaborative agents, and LLM-as-a-judge for slate recommendation.

These tools are useful, but they should not replace online experiments. They are best treated as intermediate gates: useful for catching invalid outputs, unsafe explanations, poor diversity, or obvious preference mismatch before exposing users to a system. Online A/B remains the standard for business impact, and long-term holdouts remain necessary for retention and ecosystem health.

14 Deployment Patterns

14.1 Offline large-model artifact generation

The safest deployment pattern is offline artifact generation. LLMs generate tags, semantic IDs, intent summaries, user profiles, graph edges, or synthetic samples. These artifacts are validated, cached, and used by conventional rankers. ReLand, LLM-KERec, AIR, AKT-Rec, and many feature-engineering papers follow this route.

14.2 Nearline and incremental inference

Nearline inference updates features or candidates every few minutes or hours. SIGMA aggregates users' latest behaviors at minute-level intervals and uses incremental inference with cached KV states. This pattern improves freshness without putting full LLM inference into every request.

14.3 Candidate-pool augmentation

PLUM adds generated candidates to the existing YouTube retrieval pool rather than replacing the whole retrieval stack. This is a common production strategy: introduce a new retriever as one channel, control its quota, and let downstream rankers and guardrails determine final exposure.

14.4 On-device intent agents

RecGPT-Mobile points to a privacy-preserving future: a small on-device LLM summarizes immediate context or predicts the next query, while the cloud handles global retrieval and ranking. This reduces server cost and privacy risk, but creates model update, device heterogeneity, and safety challenges.

14.5 Full large recommender models

HSTU and MTFM show that large-scale recommendation-native models can produce online gains. They require deep systems work: attention variants, feature tokenization, micro-batching, cache reuse, subgraph serving, distributed training, and monitoring. These are not simple LLM API integrations.

15 Risks and Failure Modes

- **Semantic-collaborative mismatch.** A language model may recommend semantically plausible items that users do not click or buy.
- **Calibration failure.** LLM probabilities or token scores are not calibrated CTR/CVR/GMV estimates.
- **Invalid generation.** Generated item IDs, semantic IDs, or titles may not map to available catalog items.
- **Popularity and exposure bias.** LLMs may reproduce common web popularity or platform exposure patterns.
- **Length and position bias.** Longer item descriptions, longer prompts, or earlier candidate positions can receive unfair advantage.
- **Privacy leakage.** User summaries and explanations may expose sensitive behavior.
- **Safety and policy drift.** Generated explanations or query rewrites can violate policy even when the underlying item is safe.
- **Staleness.** Offline semantic artifacts can become stale when products, trends, or user contexts change.
- **Cost opacity.** Offline AUC gains can disappear when model latency, GPU cost, or index refresh cost is included.

16 A Practical Playbook

For a team building an LLM-enhanced recommender, the evidence suggests the following sequence.

1. Identify the bottleneck: cold start, long tail, cross-domain transfer, content understanding, search-query generation, slate diversity, explanation, or long-history compression.
2. Choose the large-model role: artifact generator, encoder, scorer, candidate generator, reranker, judge, reward model, simulator, or agent.

3. Convert LLM output into recommender-native objects: embeddings, semantic IDs, item tags, graph edges, generated candidates, intent memories, or reward features.
4. Keep a strong production baseline and introduce the large-model component as an augmenting channel first.
5. Validate invalid-output rate, hallucination, policy safety, calibration, diversity, and latency before A/B.
6. Run online A/B with target metrics and guardrails. Segment long-tail, cold-start, new users, heavy users, creators/merchants, and safety-sensitive categories.
7. Preserve long-term holdouts or delayed metrics for retention, satisfaction, and ecosystem health.

17 Open Problems

Unified reporting. Public industrial papers should report traffic split, duration, baseline maturity, target metrics, guardrails, latency, cost, and segment effects. Without this, A/B results are hard to interpret.

Semantic cache freshness. Offline LLM artifacts are cheap at serving time but can become stale. Systems need freshness-aware semantic caches and update policies.

Score calibration. Generative likelihoods, LLM judgments, and semantic rewards must be mapped to business-calibrated scores.

Constrained decoding at scale. Generative retrieval needs fast prefix constraints, validity checks, and collision handling for billion-item catalogs.

Long-term value. Short-term CTR can conflict with retention, trust, merchant fairness, creator health, and content diversity.

On-device recommendation. Local intent agents may improve privacy and immediacy, but they introduce device resource limits and distributed safety concerns.

LLM-based evaluation. LLM judges can expand labels and simulate users, but they must be calibrated against real behavior and not become circular evaluators of LLM recommenders.

18 Conclusion

The field is no longer about whether LLMs can recommend. They can, but raw LLM recommendation is rarely the industrial answer. The stronger conclusion is that large models become useful when their semantic, generative, multimodal, or reasoning capabilities are converted into controllable recommender-system objects. The most convincing public deployments either augment the existing stack with LLM-derived artifacts or build recommendation-native large models with careful serving design. Academic work continues to expand the design space, especially in semantic IDs, collaborative alignment, agentic interaction, fairness, and evaluation. The next major progress will likely come from systems that combine these ideas: foundation-scale user/action models, grounded semantic IDs, real-time intent memory, calibrated business objectives, and evaluation protocols that measure long-term ecosystem value rather than only short-term clicks.

A Paper and System Map

This appendix is a structured map rather than an exhaustive bibliography. It groups more than two hundred related works, systems, and technical reports by their primary contribution. Many works belong to multiple categories; they are placed where they are most useful for navigation.

A.1 Classical and Pre-LLM Neural Recommendation Foundations

Cluster	Representative works	Why they matter
Industrial retrieval and ranking	Wide&Deep; YouTube DNN; Monolith; PinnerFormer; ComiRec; DSSM; TDM/JTM	DeepFM; DLRM; PinnerSage; MIND; TwinBERT; Define multi-stage retrieval/ranking, sparse feature interaction, large embedding tables, and serving constraints that LLM methods must integrate with.
Interest modeling	DIN; DIEN; DSIN; ETA; MIMN; M2GRL; MIND; HPMN	SIM; BST; Establish target attention, evolving interests, session dynamics, and long-history compression before LLMs.
Sequential recommendation	GRU4Rec; Caser; Rec; BERT4Rec; CL4SRec; DuoRec; FEARec; RecFormer	SAS-Rec; S3-Rec; FDSA; Provide the transformer sequence modeling foundation that later generative recommenders scale.
Multimodal and content recommendation	VBPR; MMGCN; LATTICE; MM-Rec; Rec; PMMRec; CLIP4Rec; VIP5	GRCN; MISS-Rec; Show how images, text, audio, and metadata support cold-start and cross-domain transfer.
Knowledge and graph recommendation	RippleNet; KGAT; KGCN; LightGCN; Sage; GraphRec; KG-enhanced conversational recommenders	KGIN; PinSage; HKFR; Prefigure LLM knowledge extraction and graph augmentation.
Conversational recommendation	ReDial; TG-ReDial; CR-Walker; UniCRS; TCP; C2-CRS; Chat-REC	KBRD; MESE; Provide interaction protocols and evaluation concerns for agentic recommendation.

A.2 LLM for Feature Engineering and Data Augmentation

Subtopic	Representative works	Summary
User/item feature augmentation	LLM4KGC; TagGPT; PIE; LGIR; HKFR; LLaMA-E; CUP; SINGLE; LLMHG; Llama4Rec; LoRec	ICPC; KAR; LLM-Rec; EcomGPT; TF-DCon; RLMRec; LLMRec; LLMRG; UEM; SAGCN; LLM4Vis; LLMs extract tags, aspects, user interests, knowledge, and concise profiles. The output is usually cached and passed to conventional models.
Sample and query generation	GReaT; DPLLM; RecPrompt; Agent4Ranking; ONCE; MINT; PO4ISR; PopNudge; RecInter	AnyPredict; Agent4Rec; BEQUE; LLMs generate synthetic interactions, narrative preferences, long-tail query rewrites, or simulated interactions.
Knowledge extraction	KAR; KERec; LlamaRec-LKG-RAG; DOKE	TRAWL; HKFR; ReLand; LLMRG; LKPNR; The LLM is used as a knowledge producer rather than a direct ranker.
Industrial artifact generation	ReLand; Rec; Taiji; Mobile	LLM-KERec; AIR; AKT-Rec; RecGPT; RecGPT-Generate or compress semantic artifacts offline/nearline, then serve with standard retrieval/ranking infrastructure.

A.3 LLM and PLM Encoders for Recommendation

Subtopic	Representative works	Summary
Text-rich item encoding	U-BERT; UNBERT; PLM-NR; Pyramid-ERNIE; ERNIE-RS; CTR-BERT; SuKD; PREC; Tiny-NewsRec; PLM4Tag; TwHIN-BERT; RecFormer; TBIN; Social-LLM	Encoders improve item/user representation when metadata and text are central.
Cross-domain and transferable encoders	ZESRec; UniSRec; TransRec; VQ-Rec; IDRec vs MoRec; S&R Foundation; MISSRec; UFIN; PMMRec; Uni-CTR; PCDR	Use language or multimodal embeddings to transfer across domains and cold-start scenarios.
LLM embeddings and adapters	LLM2BERT4Rec; LLM4ARec; UEM; CollabContext; SSNA; KERL; LLMRS; LLMHG; HLLM; HLLM-style item/user hierarchies	Use frozen embeddings, adapters, or hierarchical LLMs to represent user histories and item content.
Industrial encoders	PLUM item semantic representations; AKT-Rec MLLM semantic clusters; MTFM heterogeneous tokens; HSTU action tokens	The encoder is redesigned around production data streams, not only natural language.

A.4 LLM Scoring, Ranking, and Instruction Tuning

Subtopic	Representative works	Summary
Prompted scoring and rankers	LMRecSys; Zero-shot GPT; BookGPT; PALR; LLMRanker; SetwiseRank; LiT5; LlamaRec; LLM4Rerank; LLM as explainable re-ranker	Use LLMs to score or rank candidate sets; strong for analysis and small candidate pools, costly for full ranking.
Instruction tuning	TALLRec; InstructRec; GenRec; BIGRec; DEALRec; RecRanker; Recommendation as Instruction Following; Aligning LLMs with Recommendation Knowledge; RDRec	Align LLMs to recommendation labels and tasks through instructions, LoRA, or distillation.
Collaborative alignment	CoLLM; LLaRA; A-LLMRec; CLLM4Rec; LC-Rec; ControlRec; FLIP; ClickPrompt; Llama4Rec; iLoRA; CALRec	Inject ID/collaborative signals into language models to reduce semantic-collaborative mismatch.
Retrieval-augmented ranking	ReLLa; RALLRec; LlamaRec-LKG-RAG; CRAG; retrieval-augmented conversational recommendation	Retrieve relevant user history, knowledge, or candidates before LLM scoring.
Incremental and efficient tuning	POD; LSAT; E4SRec; LightLM; MoLoRec/RecCocktail; instruction distillation; decoding acceleration frameworks	Reduce compute, improve incremental updates, and make LLM rankers more deployable.

A.5 Generative Recommendation and Semantic IDs

Subtopic	Representative works	Summary
Task unification	P5; M6-Rec; InstructRec; VIP5; UP5; P5-ID; RecSysLLM; UPSR	Convert multiple recommendation tasks into language modeling or instruction-following tasks.

Subtopic	Representative works	Summary
Semantic-ID retrieval	re- TIGER; P5-ID; How to Index Item IDs; LC-Rec; IDGenRec; LETTER; Semantic IDs for Joint Search and Recommendation; GenCDR; GLASS; DaV-Gen	Generate item identifiers rather than scores; aims at unified retrieval and generation.
Semantic-ID agnostics and improvements	di- Generative Recommendation with Semantic IDs handbook; Differentiable Semantic ID; Variable-Length Semantic IDs; Generating Long Semantic IDs in Parallel; SIDInspector; Qualification-Aware SID; Adaptive Collision Handling; Expressiveness Limits; LBR	Address collision, invalid outputs, length bias, scalability, and calibration.
Industrial semantic-ID systems	sys- PLUM; SIGMA; AKT-Rec; AIR intent IDs; Taiji semantic features; Gryphon; YouTube SemanticID discussions	Show how semantic IDs enter real retrieval and ranking systems.
Diffusion and alternative generation	Diffusion recommendation surveys; continuous-token diffusion; generative recommendation; POI semantic-ID generation; chain-of-recommendation	Broaden generation beyond autoregressive text/ID decoding.

A.6 Agentic, Conversational, and Interactive Recommendation

Subtopic	Representative works	Summary
Conversational recommendation	TG-ReDial; KBRD; UniCRS; MESE; Chat-REC; Large Language Models as Zero-Shot Conversational Recommenders; retrieval-augmented CRS	Natural-language preference elicitation and explanation.
Agents for recommendation	Agent4Rec; AgentCF; MACRec; RecMind; iAgent; MMEACR; connected-TV agentic recommendation; group recommender LLM agents; controllability-centric collaborative agents	Use planning, memory, reflection, tool calls, and multi-agent collaboration.
User simulation and A/B agents	A/B Agent; RecInter; generative agents in recommendation; LLM preference induction; LLM-as-a-judge slate recommendation	Simulate users, expand labels, or evaluate slates before online tests.
Industrial agentic systems	RecGPT; RecGPT-V2; RecGPT-Mobile; Ranking Engineer Agent style ads tooling; creator/advertiser matching with Gemini-like models	Use agents for intent reasoning, engineering productivity, or matching workflows.

A.7 Trustworthy, Fair, Robust, and Safe LLM Recommendation

Subtopic	Representative works	Summary
Fairness and bias	FaiRLLM; RECLLM; job recommendation demographic bias; fairness in LLM-based recommender systems survey; provider fairness in news recommendation; popularity bias studies	Study demographic, provider, exposure, recency, and popularity biases.
Robustness and poisoning	LoRec; STELLA; exact and efficient incremental learning; memorization behavior in generative recommendation	Address instability, privacy, poisoning, and memorization.
Explainability	XRec; RecExplainer; explanation generation surveys; uncertainty-aware explainable recommendation; coherent natural-language explanations	LLMs can explain, but explanations must be faithful and safe.
Privacy and on-device	DPLLM; RecGPT-Mobile; federated content representation; privacy-preserving local LLM recommenders	Reduce central exposure of user histories and sensitive profiles.
Evaluation and controllability	LLM-as-a-judge; user preference induction; controllability-centric evaluation; A/B Agent; length bias reduction; position-bias-aware reranking	Evaluate behavior beyond static offline labels.

B Extended Annotated Literature Catalog

The previous maps group works by theme. Table 9 expands the map into an annotated catalog. It is intentionally broad: it includes classical recommenders, pre-trained language model recommenders, LLM-enhanced recommenders, generative recommenders, industrial systems, evaluation papers, fairness and robustness studies, and recent 2026 preprints. The goal is not to claim that all works are equally mature, but to show how the research frontier and production practice connect.

Table 9: Annotated catalog of representative recommendation and large-model works.

Area	Work or system	Period	Main contribution or relevance
Classical rec	deep Wide & Deep	2016	Combines memorization and generalization for industrial app recommendation.
Classical rec	deep YouTube DNN	2016	Defines large-scale candidate generation and ranking for video recommendation.
Classical rec	deep DeepFM	2017	Learns feature interactions for CTR prediction without manual cross features.
Classical rec	deep NCF	2017	Neural collaborative filtering baseline for user-item interaction modeling.
Classical rec	deep DIN	2018	Introduces target attention over user behavior for CTR prediction.
Classical rec	deep DIEN	2019	Models interest evolution with sequential behavior dynamics.
Classical rec	deep DSIN	2019	Captures session-level user interest for CTR prediction.
Classical rec	deep MIND	2019	Represents multiple user interests for large-scale retrieval.
Classical rec	deep ComiRec	2020	Extracts multiple interests for controllable candidate generation.
Classical rec	deep SIM	2020	Retrieves long user behavior sequences for target-aware modeling.
Classical rec	deep ETA	2021	Efficient target attention over long histories for industrial ranking.
Classical rec	deep DLRM	2019	Standardizes sparse embedding and dense interaction architecture at Meta scale.

Area	Work or system	Period	Main contribution or relevance
Classical rec	deep Monolith	2022	ByteDance online training system for collisionless real-time recommendation.
Classical rec	deep PinnerSage	2018	Pinterest graph-based candidate generation at web scale.
Classical rec	deep PinnerFormer	2022	Transformer user representation for Pinterest recommendations.
Classical rec	deep TDM	2018	Tree-based deep model for efficient large-corpus retrieval.
Classical rec	deep Joint tree-model optimization	2019	Jointly learns tree index and deep retrieval model.
Sequential rec	GRU4Rec	2015	Early neural session recommendation with recurrent networks.
Sequential rec	Caser	2018	Convolutional sequence embedding for next-item recommendation.
Sequential rec	SASRec	2018	Self-attentive sequential recommendation, foundation for later transformer rec.
Sequential rec	BERT4Rec	2019	Bidirectional masked-item modeling for sequential recommendation.
Sequential rec	S3-Rec	2020	Self-supervised pretraining tasks for sequential recommendation.
Sequential rec	CL4SRec	2021	Contrastive learning for robust sequential recommendation.
Sequential rec	DuoRec	2022	Contrastive regularization for sequential recommendation.
Sequential rec	FDSA	2019	Feature-level self-attention for sequential recommendation.
Sequential rec	FEARec	2023	Frequency-domain enhanced sequential recommendation.
Sequential rec	RecFormer	2023	Text-rich sequential recommendation with language representations.
Sequential rec	HSTU	2024	Recommendation-native sequential transducer that scales to trillion parameters.
PLM rec	U-BERT	2021	Pretrains user representations with BERT-style objectives.
PLM rec	UNBERT	2021	User-news matching with BERT for news recommendation.
PLM rec	PLM-NR	2021	Uses pre-trained language models to represent news content.
PLM rec	Pyramid-ERNIE	2021	Industrial Baidu search ranking with pre-trained language modeling.
PLM rec	ERNIE-RS	2021	Web-scale retrieval with ERNIE pretraining.
PLM rec	CTR-BERT	2021	Distills large BERT teacher models for CTR prediction.
PLM rec	SuKD	2022	Supplements sponsored search CTR models with NLP features.
PLM rec	PREC	2022	Plug-and-play pretraining for news recommendation.
PLM rec	Tiny-NewsRec	2022	Efficient PLM-based news recommendation.
PLM rec	TwHIN-BERT	2022	Socially enriched multilingual tweet representations.
PLM rec	Text Is All You Need	2023	Shows language representations can support sequential recommendation.
Cross-domain rec	ZESRec	2021	Zero-shot recommendation via transferable semantic representations.
Cross-domain rec	UniSRec	2022	Universal sequence representation learning for recommender systems.
Cross-domain rec	VQ-Rec	2023	Vector-quantized item representation for transferable recommendation.
Cross-domain rec	ID vs modality revisited	2023	Studies when ID-based and modality-based models win.
Cross-domain rec	Search and recommendation foundation model	2023	Unifies search and recommendation for cold-start scenarios.
Cross-domain rec	MISSRec	2023	Multi-modal interest-aware sequence pretraining.
Cross-domain rec	PMMRec	2024	Multi-modality for transferable recommender systems.
Cross-domain rec	PCDR	2024	Federated prompt-enhanced representation for cross-domain recommendation.
LLM survey	How Can RS Benefit from LLMs	2024	Surveys LLM roles across recommendation pipeline.

Area	Work or system	Period	Main contribution or relevance
LLM survey	LLMs for Recommendation Survey	2023	Early taxonomy of LLM-based recommendation.
LLM survey	Next-generation LLM-based RS	2024	Three-tier taxonomy from representation to industrial deployment.
LLM survey	LLMs for Generative Recommendation	2024	Survey of generative recommendation with LLMs.
LLM survey	LLM-enhanced RS survey	2024	Broad survey of LLM-enhanced recommendation systems.
LLM survey	Multimodal LLM recommender survey	2025	Taxonomy for LLMs in multimodal recommendation.
LLM survey	LLM agents for recommender systems	2025	Surveys agentic recommendation and search.
LLM survey	LLM fairness in recommender systems	2026	Focuses on bias mechanisms and fairness targets.
LLM feature	TagGPT	2023	Uses LLMs as zero-shot multimodal taggers.
LLM feature	LLM4KGC	2023	Few-shot relation labeling in e-commerce knowledge graphs.
LLM feature	User Interest Journeys	2023	Uses LLMs to model user interest journeys.
LLM feature	KAR	2023	Knowledge augmentation from LLMs for open-world recommendation.
LLM feature	PIE	2023	Product information extraction using ChatGPT.
LLM feature	LGIR	2023	LLM-based generative adversarial networks for job recommendation.
LLM feature	GIRL	2023	Generative job recommendations with large language models.
LLM feature	LLM-Rec prompting	2023	Personalized recommendation through prompting LLMs.
LLM feature	HKFR	2023	Heterogeneous knowledge fusion via LLMs.
LLM feature	LLaMA-E	2023	E-commerce authoring with multi-aspect instruction following.
LLM feature	EcomGPT	2023	E-commerce instruction tuning with chain-of-task tasks.
LLM feature	RLMRec	2023	Representation learning with LLM-generated knowledge.
LLM feature	LLMRec graph augmentation	2023	Uses LLMs and graph augmentation for recommendation.
LLM feature	LLM reasoning graphs	2023	Builds reasoning graphs for recommender systems.
LLM feature	CUP	2023	Concise user profiles generated from review text.
LLM feature	SINGLE	2023	Models article viewing flow using LLM-derived preferences.
LLM feature	SAGCN	2023	Semantic aspect-aware review exploitation.
LLM feature	LLMHG	2024	LLM-guided hypergraph learning for explainable recommendation.
LLM data	GReaT	2023	Language models as realistic tabular data generators.
LLM data	ONCE	2023	Boosts content recommendation with open and closed LLMs.
LLM data	AnyPredict	2023	Foundation model for tabular prediction.
LLM data	DPLLM	2023	Differentially private synthetic query generation.
LLM data	MINT	2023	Narrative-driven recommendations with LLMs.
LLM data	Agent4Rec	2023	Generative agents for dynamic recommendation simulation.
LLM data	RecPrompt	2023	Prompt tuning framework for news recommendation.
LLM data	PO4ISR	2023	Intent-driven session recommendation with LLMs.
LLM data	BEQUE	2023	LLM-based long-tail query rewriting in Taobao search.
LLM data	Agent4Ranking	2023	Multi-agent personalized query rewriting for ranking.
LLM scoring	Zero-shot recommendation as language modeling	2021	Early GPT-style zero-shot next-item framing.
LLM scoring	PTab	2022	Pre-trained language model for tabular data.
LLM scoring	UniTRec	2023	Text-to-text transformer with contrastive learning.
LLM scoring	Prompt4NR	2023	Prompt learning for news recommendation.
LLM scoring	FLAN-T5 rating prediction	2023	Evaluates whether LLMs understand user ratings.
LLM scoring	BookGPT	2023	LLM framework for book recommendation.
LLM scoring	TALLRec	2023	Efficient LLaMA-LoRA tuning for preference prediction.
LLM scoring	PromptRec	2023	Personalized cold-start recommendation with prompts.
LLM scoring	GLRec	2023	LLM graph understanding for online job recommendation.
LLM scoring	BERT4CTR	2023	Combines PLM with non-textual CTR features.

Area	Work or system	Period	Main contribution or relevance
LLM scoring	ReLLa	2024	Retrieval-enhanced LLMs for lifelong behavior comprehension.
LLM scoring	TASTE	2023	Text matching to reduce popularity bias.
LLM scoring	ClickPrompt	2024	Uses CTR models as prompt generators for PLMs.
LLM scoring	SetwiseRank	2023	Efficient setwise zero-shot ranking with LLMs.
LLM scoring	LLMRanker	2023	Fine-grained relevance labels for zero-shot rankers.
LLM scoring	CoLLM	2024	Integrates collaborative embeddings into LLMs.
LLM scoring	FLIP	2023	Fine-grained alignment between ID models and PLMs.
LLM scoring	CLLM4Rec	2024	Collaborative LLM for recommender systems.
LLM scoring	RecExplainer	2023	Aligns LLMs for recommendation interpretability.
LLM scoring	E4SRec	2023	Efficient extensible LLM solution for sequential recommendation.
LLM scoring	LSAT	2023	Incremental learning for LLM-based recommender systems.
LLM scoring	DEALRec	2024	Data-efficient fine-tuning for LLM recommendation.
LLM generation	P5	2022	Unifies recommendation tasks as language processing.
LLM generation	M6-Rec	2022	Open-ended industrial generative recommender model.
LLM generation	GPT4Rec	2023	Generative framework for recommendation and interest interpretation.
LLM generation	VIP5	2023	Multimodal foundation model for recommendation.
LLM generation	P5-ID	2023	Studies item ID indexing for recommendation foundation models.
LLM generation	PALR	2023	Personalization-aware LLMs for recommendation.
LLM generation	ChatGPT zero-shot ranker	2023	Evaluates ChatGPT for ranking candidates.
LLM generation	AGR	2023	Early artificial general recommender experiments.
LLM generation	GPTRec	2023	GPT-style generative sequential recommendation.
LLM generation	GenRec	2023	LLM for generative recommendation.
LLM generation	UP5	2023	Fairness-aware foundation model for recommendation.
LLM generation	BIGRec	2023	Bi-step grounding paradigm for LLM recommendation.
LLM generation	KP4SR	2023	Knowledge prompt tuning for sequential recommendation.
LLM generation	POD	2023	Prompt distillation for efficient LLM recommendation.
LLM generation	RaRS	2023	Retrieval-augmented recommender system with LLMs.
LLM generation	LANCER	2023	Content-enriched language modeling for sequential recommendation.
LLM generation	TransRec	2023	Bridges LLMs and recommendation through transition modeling.
LLM generation	AgentCF	2024	Collaborative learning with autonomous language agents.
LLM generation	P4LM	2023	Factual personalized recommendation with RL.
LLM generation	LightLM	2023	Lightweight language model for generative recommendation.
LLM generation	LlamaRec	2023	Two-stage LLM ranking for recommendation.
LLM generation	LC-Rec	2023	Integrates collaborative semantics into LLM recommendation.
LLM generation	ControlRec	2023	Bridges semantic gap between LM and personalized recommendation.
LLM generation	LLaRA	2024	Large language-recommendation assistant.
LLM generation	DRDT	2023	Dynamic reflection with divergent thinking.
LLM generation	STELLA	2023	Shows LLMs are not stable recommender systems.
Semantic ID	TIGER	2023	Generative retrieval with semantic IDs.
Semantic ID	How to Index Item IDs	2023	Studies ID design for recommendation foundation models.
Semantic ID	Better Generalization with Semantic IDs	2023	Case study in ranking with semantic IDs.
Semantic ID	LMIndexer	2023	Language models as semantic indexers.
Semantic ID	LETTER	2024	Learnable item tokenization for generative recommendation.
Semantic ID	Unifying generative and dense retrieval	2024	Bridges dense retrieval and generative decoding.
Semantic ID	Inductive generative recommendation	2024	Uses retrieval-based speculation for inductive recommendation.
Semantic ID	Generating long semantic IDs in parallel	2025	Parallelizes prediction of long semantic IDs.
Semantic ID	Semantic IDs for joint search and recommendation	2025	Unifies search and recommendation with semantic IDs.

Area	Work or system	Period	Main contribution or relevance
Semantic ID	Practitioner handbook for semantic IDs	2025	Benchmarking and practical guidance for SID systems.
Semantic ID	HiD-VAE	2025	Hierarchical disentangled semantic IDs.
Semantic ID	Differentiable Semantic ID	2026	Learns SIDs end-to-end for generative recommendation.
Semantic ID	Variable-Length Semantic IDs	2026	Adapts SID length to item complexity.
Semantic ID	GenCDR	2025	Adaptive semantic tokenization for cross-domain recommendation.
Semantic ID	GLASS	2026	SID-tier and semantic search for long sequences.
Semantic ID	Gryphon	2026	Joint SID generation and item-level scoring in industry.
Semantic ID	SIDInspector	2026	Diagnostics for semantic-ID tokenizers.
Semantic ID	LBR	2026	Mitigates length bias in LLM-based recommendation.
Large rec model	HSTU	2024	Trillion-parameter sequential transducer with online A/B evidence.
Large rec model	Scaling recommender transformers to one billion parameters	2025	Studies transformer scaling for recommendation.
Large rec model	Large User Model	2025	Unlocks scaling law in industrial recommendation.
Large rec model	MTFM	2026	Meituan multi-scenario foundation model.
Large rec model	RecBase	2025	Generative foundation model pretraining for zero-shot recommendation.
Large rec model	OneMall	2026	One-model family for Kuaishou e-commerce recommendation.
Large rec model	GPR	2025	Generative pretrained one-model paradigm for ads recommendation.
Industrial A/B	ReLand	2024	Alipay controllable LLM reasoning pool with CTR and CVR gains.
Industrial A/B	LLM-KERec	2024	Alipay inferential knowledge graph with conversion and GMV gains.
Industrial A/B	HSTU GRs	2024	Meta sequential transducers with 12.4 percent online lift.
Industrial A/B	SORT-Gen	2025	Taobao generative re-ranking with click and GMV gains.
Industrial A/B	PLUM	2025	YouTube semantic-ID generative retrieval with live experiment gains.
Industrial A/B	RecGPT	2025	Taobao intent-centric reasoning model with IPV and CTR gains.
Industrial A/B	RecGPT-V2	2025	Taobao multi-agent intent reasoning with GMV and novelty gains.
Industrial A/B	MTFM	2026	Meituan foundation model with CTR, conversion, orders, and latency gains.
Industrial A/B	SIGMA	2026	AliExpress generative multi-task recommender with GMV gains.
Industrial A/B	AIR	2026	Kuaishou atomic intent reasoning with GMV and throughput gains.
Industrial A/B	Taiji	2026	Kuaishou ads LLM-enhanced recommendation with revenue gains.
Industrial A/B	AKT-Rec	2026	Tmall long-tail semantic-ID recommendation with CTR and GMV gains.
Industrial proto	BEQUE	2023	Taobao long-tail query rewriting with LLMs.
Industrial proto	Universal LLM retriever	2025	Industrial-scale multi-objective retrieval with LLMs.
Industrial proto	RecGPT-Mobile	2026	On-device intent agent for Taobao feed recommendation.
Industrial proto	AIGQ	2026	Taobao generated query architecture for e-commerce query recommendation.
Industrial proto	RecGPT explanation generation	2025	Dynamic explanations and judge/reward framework.
Industrial proto	YouTube creator matching with Gemini	2026	Product blog example of multimodal matching for brand-creator discovery.
Industrial proto	Meta GEM discussions	2025	Ads foundation-model direction; public non-primary summaries require caution.
Multimodal rec	MM-Rec	2022	Visiolinguistic multimodal news recommendation.
Multimodal rec	VIP5	2023	Multimodal foundation model for recommendation.
Multimodal rec	MISSRec	2023	Multi-modal interest-aware transfer.
Multimodal rec	PMMRec	2024	Transferable recommendation through multimodality.
Multimodal rec	MMREC	2024	LLM-based multimodal recommender system.
Multimodal rec	AKT-Rec	2026	MLLM semantic clusters for long-tail e-commerce recommendation.

Area	Work or system	Period	Main contribution or relevance
Multimodal rec	MMEACR	2026	Multimodal memory-enhanced agent collaboration.
Conversational rec	ReDial	2018	Benchmark for conversational movie recommendation.
Conversational rec	TG-ReDial	2020	Topic-guided recommendation dialogues.
Conversational rec	KBRD	2019	Knowledge graph enhanced conversational recommendation.
Conversational rec	UniCRS	2022	Unified conversational recommender system.
Conversational rec	Chat-REC	2023	Interactive and explainable LLM-augmented recommendation.
Conversational rec	Large LMs as zero-shot conversational recommenders	2023	Tests LLMs in conversational recommendation.
Conversational rec	Retrieval-augmented CRS	2024	Prompt-based state tracking with retrieval.
Agentic rec	Agent4Rec	2023	Generative agents for recommendation experiments.
Agentic rec	AgentCF	2024	Autonomous agents for collaborative filtering.
Agentic rec	MACRec	2024	Multi-agent collaboration for recommendation.
Agentic rec	RecMind	2024	Multi-step LLM reasoning and tools for recommendation.
Agentic rec	iAgent	2025	LLM agent as a shield between user and recommender.
Agentic rec	RecGPT-V2	2025	Hierarchical multi-agent system for intent reasoning.
Agentic rec	CTV agentic recommendation	2026	Agentic recommendation for connected TV content discovery.
Agentic rec	Group recommender LLM modeling	2026	Models consensus and dissent in group preferences.
Evaluation	LLMRec benchmark	2023	Benchmarks LLMs on recommendation tasks.
Evaluation	Tapping LLM potential	2024	Framework and empirical analysis of LLM recommenders.
Evaluation	A/B Agent	2026	Simulates multimodal user behavior for A/B testing.
Evaluation	LLM-as-a-judge slate recommendation	2025	Uses LLMs as world models or judges for slates.
Evaluation	User preference induction	2026	LLM expands offline top-N relevance judgments.
Evaluation	Controllability-centric evaluation	2026	Collaborative agents evaluate controllability.
Evaluation	Powerful A/B metrics	2024	Metric design for recommender online experiments.
Fairness	FaiRLLM	2023	Evaluates fairness in LLM recommendation.
Fairness	UP5	2023	Unbiased foundation model for fairness-aware recommendation.
Fairness	ChatNews provider fairness	2023	Studies personalization, fake news, and provider fairness.
Fairness	Demographic bias in job recommendation	2023	Shows unequal opportunities in LLM job recommendation.
Fairness	RECLLM	2024	Studies provider fairness, stability, and recency.
Fairness	Fairness in LLM4Rec survey	2026	Systematic review of bias mechanisms and mitigation.
Robustness	LoRec	2024	Robust sequential recommendation against poisoning attacks.
Robustness	STELLA	2023	Demonstrates instability in LLM recommenders.
Robustness	Exact unlearning for LLM recommendation	2024	Machine unlearning for LLM recommenders.
Robustness	Memorization in generative recommendation	2026	Studies memorization behavior of LLM generative recommenders.
Robustness	LBR	2026	Mitigates input and output length bias.
Explanation	XRec	2024	LLMs for explainable recommendation.
Explanation	RecExplainer	2023	Aligns LLMs to explain recommendation models.
Explanation	Coherence in explanations	2023	Studies natural-language explanation coherence.
Explanation	Review of LLM explanations	2024	Surveys recommendation explanation with LLMs.
Privacy	DPLLM	2023	Differentially private synthetic query generation.
Privacy	Federated prompt content representation	2024	Federated cross-domain content representation.
Privacy	RecGPT-Mobile	2026	Device-side LLM intent understanding for privacy and freshness.
Ads and commerce	SuKD	2022	NLP features for sponsored search CTR.
Ads and commerce	BERT4CTR	2023	PLM and non-text features for CTR prediction.

Area	Work or system	Period	Main contribution or relevance
Ads and commerce	Uni-CTR	2023	Multi-domain CTR prediction via LLMs.
Ads and commerce	Taiji	2026	LLM-enhanced ads recommendation with ADVV and revenue gains.
Ads and commerce	RecGPT	2025	Taobao large reasoning model for homepage recommendation.
Ads and commerce	AIGQ	2026	E-commerce generated query recommendation.
News rec	PLM-NR	2021	PLM-empowered news recommendation.
News rec	Tiny-NewsRec	2022	Efficient news recommendation.
News rec	Prompt4NR	2023	Prompt learning for news recommendation.
News rec	RecPrompt	2023	Prompt tuning for news recommendation.
News rec	LLM-based news recommender survey	2025	Domain-specific survey for news recommendation.
POI and local rec	Generative next POI with semantic ID	2025	Semantic-ID generation for next POI recommendation.
POI and local rec	BTRec	2023	BERT-based trajectory recommendation for tours.
POI and local rec	Tour itinerary recommendation with LMs	2023	Language models for travel itinerary recommendation.
Search and rec	S and R foundation model	2023	Unified search and recommendation for cold start.
Search and rec	Semantic IDs for joint search and recommendation	2025	Shared SID design for both tasks.
Search and rec	AIGQ	2026	Hybrid generated query recommendation in Taobao.
Search and rec	BEQUE	2023	Long-tail query rewriting.
Search and rec	MUVERA-like multi-vector retrieval discussions	2025	Adjacent retrieval technology relevant to LLM-era recommendation.
Recent 2026	AIR	2026	LLM atomic intent reasoning for cross-domain commerce recommendation.
Recent 2026	Taiji	2026	Pareto optimization of semantic and ID rewards.
Recent 2026	MTFM	2026	Multi-scenario foundation model in Meituan.
Recent 2026	SIGMA	2026	Instruction-driven generative multi-task recommender.
Recent 2026	AKT-Rec	2026	Long-tail semantic-ID transfer at Tmall.
Recent 2026	LBR	2026	Length bias reduction for LLM recommendation.
Recent 2026	MMEACR	2026	Multimodal memory agents for recommendation.
Recent 2026	Connected TV agent	2026	LLM-powered CTV content discovery.
Recent 2026	User preference induction	2026	LLM-based offline evaluation label expansion.
Recent 2026	Controllability evaluation	2026	Collaborative agents evaluate steering and controllability.
Recent 2026	Consensus versus dissent group rec	2026	LLM modeling of subjective group preferences.
Recent 2026	Intent-driven SID news recommendation	2026	Grounded conversational news recommendation with semantic IDs.

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